

Student Pre Post Retrospective

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## Warning: package 'surveydown' was built under R version 4.6.1
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— consentpage

Instructions to Participants

Thank you for participating in the survey! We are collecting data about your experience learning genomics and genetics topics. This data will be used to guide the AI-GENe Program and may ultimately be used for a research publication and reporting to the NIH.

Data confidentiality

Responses are anonymous. The potential risks to you are small. The potential benefits to the education community are high—we will use this information to guide development of education material. Please do not include names, institution names, email addresses, or other identifying information in your written responses.

Data sharing

No personally identifiable information, such as response timestamps, will be shared.

Participation

You must be 18 or over to participate. You will be asked a series of questions about participating in AI-GENe modules. **By completing this survey or questionnaire, you are consenting to be in this research study. Your participation is voluntary, and you can stop at any time.** You do not need to answer optional questions. Any answers you have filled in will be automatically submitted if you close the browser tab.

Depending on how you answer the survey, this will take about **3–6 minutes** to complete. Taking this survey will not affect your grade.

Risk

Risk of participation is minimal. The questions we ask pose minimal security risk to participants, as they focus on your personal experience and expectations. While we have taken measures to ensure the security of the survey responses, please be aware that no online system is completely secure. By participating in this survey, you understand and accept the inherent risks associated with transmitting information over the internet.

Consent

The first question asks whether you are at least 18 years old and whether you consent, or agree, that data from your responses may be used for AI-GENe development purposes.

Questions or problems

If you have questions or problems, contact:

- Ava Hoffman at ahoffma2@fredhutch.org (<mailto:ahoffma2@fredhutch.org>)
- Michael Schatz at mschatz@jhu.edu (<mailto:mschatz@jhu.edu>)

You may also contact the Johns Hopkins University Homewood Institutional Review Board (IRB) if you have questions about your rights as a participant. Contact the IRB if you feel you have not been treated fairly or if you have other concerns.

- **IRB No.:** HIRB00024197
- **Date approved:** XX/XX/2026
- **HIRB Office phone:** 410-516-6580
- **HIRB Office email:** hirb@jhu.edu (<mailto:hirb@jhu.edu>)

Consent for answers to be used by AI-GENe staff

I am 18 or over AND I consent to my responses being used for AI-GENe's evaluation and development.

- ☐ Yes
- ☐ No

*

— background

About your institution and module

Which of these best describes the institution where you attend?

- ☐ Doctoral University - Very high research activity ("R1")
- ☐ Doctoral University - High research activity ("R2")
- ☐ Master's College or University
- ☐ Primarily Undergraduate Institution (PUI)
- ☐ Community College
- ☐ High School
- ☐ Other

*

(Optional) If you selected "Other," please provide a description of the type of institution you attend.

*

Which AI-GENe module did you complete?

- ☐ Genome Annotation and the Phenotypic Effects of Variants
- ☐ Predicting Gene Expression Patterns
- ☐ Predicting Protein Structure from Genomic Data
- ☐ Sample Collection and HMW DNA Isolation
- ☐ Assessing and Improving DNA Quality
- ☐ DNA Library Preparation and Long Read Sequencing
- ☐ Sequencing Data QC & Taxonomic Profiling
- ☐ Metagenome Assembly & AMR Gene Annotation
- ☐ LLMs and Scientific Literacy

*

— reflection1

Reflecting on your experience

These questions are intended to help you reflect on your experience working through the AI-GENe modules. Please rate how much you agree with each statement.

After completing the activity, my skills and comfort with genetics lab work have improved.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

After completing the activity, my skills and comfort with analyzing genetic data have improved.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

After completing the activity, my skills and comfort with using genomics tools and/or websites have improved.

- ☐ Not at all (1)
- ☐ Slightly (2)

- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

After completing the activity, my skills and comfort with using AI-based tools for genetics work have improved.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

This activity made me apply my skills, rather than recall facts.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

This activity made me evaluate evidence, rather than recall facts.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

This activity helped me draw conclusions from data.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

This activity allowed me to do “real science.”

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

After completing the activity, I have more interest in STEM careers.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

After completing the activity, I am more knowledgeable about how metagenomics research happens.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

After completing the activity, I am more knowledgeable about how AI models work.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

After completing the activity, I am more curious about genomics/metagenomics/genetics.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

After completing the activity, I want to learn more about genomics/genetics.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

After completing the activity, I have a better understanding of how AI is used in genomics/metagenomics research.

- ☐ Not at all (1)
- ☐ Slightly (2)
- ☐ Moderately (3)
- ☐ Quite a bit (4)
- ☐ Very much (5)

*

(Optional) In 1–2 sentences, tell us about your responses above.

*

— openended

Additional reflections

The following open-ended questions are **optional**.

Please do not include names, institution names, email addresses, or other identifying information in your responses.

(Optional) In 1–2 sentences, tell us something you are more knowledgeable about.

*

(Optional) Were any parts of the activity particularly engaging? If so, please describe.

*

(Optional) Did you find any parts of the activity confusing? If so, please describe.

*

(Optional) What suggestions, if any, do you have for lesson improvement?

*

— end

Thank you for completing the AI-GENe survey.

Click **Exit Survey** below to finish.

Exit Survey

— ineligible

Thank you for your interest. You must be at least 18 years old and consent to the use of your responses for AI-GENe evaluation and reporting purposes to participate in this survey.

Click **Exit Survey** below to finish.

Exit Survey